

Homework (Jalapeno)

- Q1.** In ancient Egypt, a survey of 100 pharaohs found that 60 liked *sphinx · statues* and 40 liked *mummy · wraps*. If 20 pharaohs liked both, how many liked only *sphinx · statues*?
- (A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q2.** A two-way table of artifacts from the ancient city of Pompeii shows that 15 out of 25 *vases* had *glazed · finishes*. What is the relative frequency of *vases* with *glazed · finishes*?
- (A) 0.2
(B) 0.4
(C) 0.6
(D) 0.8
(E) 1.0
- Q3.** An association between *climate* and *agriculture* in ancient civilizations can be analyzed using two-way tables. If the table shows that 80
- (A) Hot climates always lead to irrigation
(B) Irrigation is more common in hot climates
(C) Hot climates have no effect on irrigation
(D) Irrigation is less common in hot climates
(E) Other factors influence irrigation more than climate
- Q4.** A two-way table of data from an archaeological site shows the number of *potsherds* found in different *strata*. If 30 *potsherds* were found in *stratum A* and 20 were found in both *stratum A* and *stratum B*, what is the number of *potsherds* found only in *stratum A*?
- (A) 10
(B) 15
(C) 20
(D) 25
(E) 30
- Q5.** The ancient Greeks used two-way tables to analyze the relationship between *olympic events* and *athlete participation*.
- If 40 athletes participated in the *javelin throw* and 20 also participated in the *discus throw*, what percentage of *javelin throw* athletes also participated in the *discus throw*?
- (A) 25
(B) 40
(C) 50
(D) 60
(E) 75
- Q6.** A museum has a collection of ancient *coins* with different *denominations*. A two-way table shows that 50 *coins* have a *denomination* of 1 *drachma* and 20 have a *denomination* of both 1 *drachma* and 2 *drachma*. How many *coins* have only a *denomination* of 1 *drachma*?
- (A) 10
(B) 20
(C) 30
(D) 40
(E) 50
- Q7.** The construction of the ancient *pyramids* required the use of *materials* like *limestone* and *granite*. A two-way table shows the number of *pyramids* built with each *material*. If 15 *pyramids* were built with *limestone* and 5 were built with both *limestone* and *granite*, what is the percentage of *pyramids* built with only *limestone*?
- (A) 33
(B) 50
(C) 60
(D) 66
(E) 75
- Q8.** A historical study of the *black death* found an association between *social class* and *survival rate*. If 60
- (A) Social class has no effect on survival rate
(B) Lower class has a higher survival rate
(C) Upper class has a higher survival rate
(D) Other factors influence survival rate more than social class
(E) The relationship is unclear

Open Ended Questions (FRQ)

- Q9.** Create a two-way table to analyze the relationship between *ancient civilization* and *writing system*. Include at least 5 civilizations and 3 writing systems. Calculate the relative frequencies for each cell in the table.
- Q10.** A two-way table shows the number of *artifacts* found in different *archaeological sites*. Analyze the association between *site location* and *artifact type*. Provide examples of how the relative frequencies can be used to draw conclusions about the relationship between these variables.

Chef's Tips

- Tip 1: Review the learning objectives for Unit 9 to ensure understanding of two-way tables
- Tip 2: Use real-world historical examples to illustrate the concepts of relative frequency and association
- Tip 3: Encourage students to use LaTeX commands to format their math responses